Elijah DiFuria

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Education

University of California, Santa Barbara — September 2023 – June 2026
B.S. in Statistics & Data Science, 3.60 GPA
Achievements: Dean's Honors Fall 2023/Winter 2024

Relevant Coursework: Statistical Machine Learning, Stochastic Process, Regression Analysis, Bayesian Statistics, Design of Experiments, Survival Analysis, Intermediate Python Programming

Projects

Wikispeedia Network Analysis — March 2026
Course Project

- Modeled a real-world Wikipedia hyperlink network (~4,600 nodes, ~120k edges) as a directed graph using R (igraph)
- Applied Louvain, Fast Greedy, and Walktrap algorithms and evaluated communities using modularity
- Visualized clustering patterns by using Fruchterman-Reingold, Kamada-Kawai, and adjacency matrices
- Performed a Monte Carlo null model (1,000 simulations) to validate statistical significance of results

Taste Perception Experiment — March 2025
Course Project

- Designed and conducted a factorial experiment on how pouring/consumption methods impact taste perception
- Collected and analyzed data using R (linear models, ANOVA, permutation tests) to assess statistical significance
- Addressed model assumption violations through diagnostic tests and permutation-based inference
- Communicated findings clearly in a structured report, emphasizing consumer insights and study limitations.

Academic Performance Analysis — June 2025
Course Project

- Built a multiple linear regression model using R to predict exam scores using lifestyle and study variables
- Performed regression diagnostics (Q–Q, residuals, Cook's distance, Shapiro-Wilk) to validate assumptions
- Generated a correlation matrix to assess relationships and identify the strongest predictors
- Produced a neatly structured report with figures, tables, and interpretable recommendations

Work Experience

UC Santa Barbara Recreation Center – Operations Assistant — June 2025 - Present

- Assisted patrons with equipment, questions, and facility use while maintaining safety and policy compliance.

Wild Waters Adventure Park – Lead Lifeguard — Summer 2022/2023

- Supervised a rotation of 5 lifeguards ensuring everyone had smooth rotations, lunch breaks, and were well-supported.
- Tasked with ensuring safety, enforcing rules, and accommodating guests.

Skills

- Python, R Studio, SQL, JavaScript, C#
- ANOVA, Binary Search Trees, Pandas, ggplot2, GitHub
- Regression Analysis, Bayesian Statistics, Survival Analysis, Statistical Machine Learning, Stochastic Processes
- Time Management, Clear Communication, Leadership, Analytical

Hobbies & Clubs

- USCB Data Science Club Project Manager, UCSB Pickleball Club Officer
- Surfing, Golfing, Weightlifting, Drumming, Piano